Geometric patterns

1

a 3

b The number of matchsticks is equal to three times the number of triangles.

c $M = 3T$

2

a

Figure number	1	2	3	4
No. matchsticks	5	6	7	8

b The number of matchsticks needed to construct a shape is equal to the entry number of that shape plus four.

c $M = F + 4$

3

a 3

b The number of matchsticks increases by three

c The number of matchsticks is equal to three times the number of squares plus one.

d $M = 3S + 1$

4

a

Entry number (E)	1	2	3	4
No. of tiles used (T)	1	4	9	16

b The number of tiles is equal to the entry number multiplied by itself.

c $T = E^2$

Equations and tables

5

a $y = x - 6$　　b $y = 3x$　　c $y = 2x + 5$

6

a

i The value for y is equal to the value for x plus eight.

ii $y = x + 8$

b

i The value for y is equal to the value for x divided by three.

ii $y = x \div 3$

c

i The value for y is equal to the value for x minus five.

ii $y = x - 5$

d

i The value for y is equal to the value for x times seven.

ii $y = 7x$

e

i The value for y is equal to the value for x times five plus two.

ii $y = 5x + 2$

f

i The value for y is equal to the value for x times minus three plus seventeen.

ii $y = -3x + 17$

7

a $y = x - 3$ b $y = 5x$ c $y = x + 7$ d $y = 2x + 2$

e $y = 2x - 8$ f $y = \frac{x}{9}$

8

a

x	1	2	3	4
y	7	14	21	28

b

x	1	2	3	4
y	9	10	11	12

c

x	1	2	3	4
y	−4	2	0	2

d

x	1	2	3	4
y	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{3}{4}$	1

e

x	1	2	3	4
y	6	11	16	21

f

x	1	2	3	4
y	−3	−6	−9	−12

g

x	1	2	3	4
y	$\frac{2}{3}$	$\frac{4}{3}$	2	$\frac{8}{3}$

h

x	1	2	3	4
y	6	2	−2	−6

Equations for relationships

9

a

Bunches, x	1	2	3	4
Bananas, y	4	8	12	16

b The number of bananas is equal to four times the number of bunches.

c $y = 4x$

10

a

Decks, d	1	2	3	4
Cards, c	52	104	156	208

b The number of cards is equal to fifty-two times the number of decks.

c $c = 52d$

11

a

Week (W)	0	1	2	3	4
Account total (A)	$300	$310	$320	$330	$340

b $A = 300 + 10W$

12

a The number of marbles will be equal to four times the number of bags plus his original five.

b $y = 4x + 5$